Teorema 1. *TFAE:*

1. *p prime*
2. *for all $\phi_1, .., \phi_n \in \Delta$ $p \vdash \bigvee_{i=1,..,n} \phi_i \Rightarrow p \vdash \phi_i$ for some $i = 1, .., n$.*

Proof.

- $1 \Rightarrow 2$: OK!
- $2 \Rightarrow 1$: It's enough showing that $p \cup \{\neg\phi : \phi \in \Delta, p \not\vdash \phi\}$ consistent. By contradiction it's inconsistent. Then let $\phi_1, .., \phi_n \in \Delta$ such that $p \cup \{\neg\phi_1, .., \neg\phi_n\}$ inconsistent and $p \vdash \neg\phi_i$ for all $i = 1, .., n$. Then $p \vdash \neg \bigwedge_{i=1,..,n} \neg\phi_i$. Then $p \vdash \bigvee \phi_i$. This implies $p \vdash \phi_i$ for some $i = 1, .., n.$, a contradiction.

□